Supplementary Material

GCZ-RL: Geometry-Calibrated Residual Learning for Spinal Ligament Landmark Localization on Vertebral Surface Meshes

Section, equation, and table numbers without the prefix S refer to the main text. Every value in this document is regenerated by the scripts of the code release (`paper/make_paper_numbers.py`; for Sections S8–S13, `paper/revision_analyses.py`, `paper/reproducibility_check.py`, and `paper/icp_seed_sensitivity.py`); the mapping from each table to its script and output file is given in `docs/PAPER_MAP.md`.

S1. Residual-regressor features

Each GCZ point $\mathbf{p}$ of ligament class L on a vertebra with vertex set $\mathcal{V}$ is described by the 36 features of Table S1. The notation is as follows. $\mathbf{v}_{\min}$ and $\mathbf{v}_{\max}$ are the componentwise minimum and maximum of $\mathcal{V}$, and $\mathbf{s} = \mathbf{v}_{\max} - \mathbf{v}_{\min}$ is the extent of the vertebra (components equal to zero are set to one). The normalized position is $\mathbf{p}_n = (\mathbf{p} - \mathbf{v}_{\min}) \oslash \mathbf{s} = (x_n, y_n, z_n)$, where $\oslash$ denotes componentwise division. $\mathcal{R}_L$ is the candidate region of class L used for projection (Section 3.4; if it contains fewer than two vertices, $\mathcal{V}$ is used), $\mathbf{c}_R$ its centroid, and $\mathbf{r}_{\min}$, $\mathbf{r}_{\max}$ its componentwise minimum and maximum. $[T_y^-, T_y^+]$ is the calibrated y_n band of class L. $\bar{\mathbf{b}}_L$ is the mean normalized expert landmark centroid of class L over the training vertebrae, and $\mathbf{b} = \mathbf{v}_{\min} + \bar{\mathbf{b}}_L \odot \mathbf{s}$ is its position on the current vertebra. $\mathcal{N}$ denotes the 30 vertices of $\mathcal{V}$ nearest to $\mathbf{p}$. All features are computed from the mesh, the GCZ point, and quantities learned from the training vertebrae; no annotation of the vertebra being corrected is used. Distances are in millimeters.

The features are produced by `point_features` and `_zone_features` in `code/ligament/gczrl.py` of the code release. Features 5 and 6 repeat features 2 and 3; they are listed because they are part of the implemented feature vector with which all reported results were obtained. For mirrored training samples (Section 3.7), the features are computed for the mirrored point on the original mesh.

Table S1: Definition of the 36 residual-regressor features.

#	Feature	Definition	Source
Position of the GCZ point			
1–3	Normalized position	x_n, y_n, z_n	Mesh bounding box
4	Distance from mid-sagittal plane	$\lvert x_n - 0.5 \rvert$	Mesh bounding box
5–6	Normalized position (repeat)	y_n, z_n	Mesh bounding box
Relation to the candidate region			
7	Distance to region centroid	$\lVert \mathbf{p} - \mathbf{c}_R \rVert_2$	Candidate region
8	Distance to region box boundary	$\min\big(\min_k(p_k - r_{\min,k}),\ \min_k(r_{\max,k} - p_k)\big)$; positive inside the bounding box of $\mathcal{R}_L$	Candidate region
9	Relative position in y_n band	$(y_n - T_y^-)/(T_y^+ - T_y^-)$; 0.5 if the band is degenerate	Calibrated thresholds
10–12	Normalized region centroid	$(\mathbf{c}_R - \mathbf{v}_{\min}) \oslash \mathbf{s}$	Candidate region
13–15	Normalized region spread	componentwise standard deviation of $\mathcal{R}_L$, divided by $\mathbf{s}$	Candidate region
16	Region size	number of vertices in $\mathcal{R}_L$	Candidate region
17–19	Region shape	singular values $\sigma_1, \sigma_2, \sigma_3$ of the centered vertices of $\mathcal{R}_L$, divided by $\sigma_1 + \sigma_2 + \sigma_3$	Candidate region
Anatomical context			
20	Vertebral level	index of the level in the order C1–C7, T1–T13, L1–L5, divided by 24	Level label
21–27	Ligament class	one-hot encoding of ALL, PLL, LF, ISL, SSL, ITL, CL	Class label
Local surface			
28–30	Surface normal	right singular vector of the smallest singular value of the centered set $\mathcal{N}$ (unit length)	Mesh, 30 nearest vertices

#	Feature	Definition	Source
31	Surface variation	$\sigma_3/(\sigma_1+\sigma_2+\sigma_3)$ of the centered set $\mathcal{N}$	Mesh, 30 nearest vertices
32	Signed lateral position	$x_n - 0.5$	Mesh bounding box
Offset from the training prior			
33	Distance to class prior	$\|\mathbf{p}-\mathbf{b}\|_2$	Training annotations (class mean)
34–36	Direction to class prior	$(\mathbf{b}-\mathbf{p}) \oslash \mathbf{s}$	Training annotations (class mean)

S2. Candidate-region construction

For class L, three normalized quantities are calibrated: y_n, $|x_n - 0.5|$, and z_n. For each, the mean μ_L and sample standard deviation σ_L of the expert landmarks of the training vertebrae, normalized within their own vertebra, give the band $[\mu_L - k_L\sigma_L,\ \mu_L + k_L\sigma_L]$ with $k_L = 2$, clipped to $[0, 1]$ (Eq. 3). The candidate region is the set of mesh vertices that lie inside all three bands. If fewer than 64 vertices remain, the z_n band is dropped and the region is formed from the y_n and $|x_n - 0.5|$ bands; in the GCZ selection step, if fewer than 64 vertices still remain, only the y_n band is used. If no vertex satisfies the bands, the whole mesh is used. These rules are implemented in `build_thresholds` and `candidate_vertices` of `code/ligament/gcz.py`.

S3. Point accounting and matching-independent metrics

The following statistics refer to the primary, patient-level leave-one-subject-out evaluation (14 lumbar vertebrae, 98 vertebra–class cases) and are computed from the saved predictions and annotations of the code release; no model was retrained. The expert annotations are dense point sets along each attachment footprint, whereas GCZ and GCZ-RL output ten landmarks per class (one for SSL and two for ITL) and SLD a class-dependent number. One-to-one Hungarian matching within each vertebra–class case pairs $\min(|P|, |G|)$ points; in every case the predicted set was the smaller one, so every predicted landmark was paired and no predicted landmark was left unpaired. No method had a missing output for any case.

Table S2: Point accounting of the primary evaluation. Ref.: number of expert reference points. Pred.: number of predicted landmarks, all of which were matched. Cov.: matched reference points as a percentage of all reference points. Unpaired reference points are Ref. minus Pred.

		GCZ		GCZ-RL		SLD	
	Ref.	Pred.	Cov.%	Pred.	Cov.%	Pred.	Cov.%
By class							
ALL	368	140	38.0	140	38.0	196	53.3
PLL	239	140	58.6	140	58.6	196	82.0
LF	349	140	40.1	140	40.1	132	37.8
ISL	330	140	42.4	140	42.4	196	59.4
SSL	14	14	100.0	14	100.0	14	100.0
ITL	28	28	100.0	28	100.0	28	100.0
CL	837	140	16.7	140	16.7	176	21.0
By held-out subject							
S01	1020	265	26.0	265	26.0	337	33.0
S02	564	212	37.6	212	37.6	264	46.8
S03	581	265	45.6	265	45.6	337	58.0
All	2165	742	34.3	742	34.3	938	43.3

Because matched-pair metrics do not penalize reference points without a prediction, Table S3 reports metrics that use every predicted and every reference point and do not depend on matching. The closest-point distance from predictions to references (P→R) is the mean, over predicted landmarks, of the distance to the nearest reference point; it is the error definition of Blomenkamp et al. (2026) and is never larger than the matched MAE. The closest-point distance from references to predictions (R→P) is the mean, over reference points, of the distance to the nearest prediction; it penalizes parts of the footprint that no prediction covers. The symmetric distance is the mean of the two, and the last block is the percentage of all reference points whose nearest prediction lies within 5 mm.

Table S3: Matching-independent metrics of the primary evaluation, averaged over the vertebra–class cases of each class; the last row averages over all 98 cases (equal to the macro average, since every class has 14 cases). Lowest distance and highest percentage per row and block in bold.

	P→R closest point (mm)			R→P closest point (mm)			Symmetric (mm)			Reference points within 5 mm (%)		
Class	GCZ	GCZ-RL	SLD	GCZ	GCZ-RL	SLD	GCZ	GCZ-RL	SLD	GCZ	GCZ-RL	SLD
ALL	2.605	2.382	**1.521**	3.989	3.798	**3.175**	3.297	3.090	**2.348**	70.5	75.4	**78.4**
PLL	1.846	1.846	**1.177**	2.329	2.329	**1.969**	2.087	2.087	**1.573**	**97.2**	**97.2**	92.3
LF	1.918	**1.912**	4.015	3.334	**3.323**	9.829	2.626	**2.618**	6.922	**80.8**	**80.8**	34.3
ISL	2.262	2.173	**1.367**	3.706	3.621	**3.143**	2.984	2.897	**2.255**	77.0	77.3	**82.1**
SSL	2.917	2.692	**1.968**	2.917	2.692	**1.968**	2.917	2.692	**1.968**	78.6	78.6	**92.9**
ITL	6.262	1.796	**0.873**	6.262	1.796	**0.873**	6.262	1.796	**0.873**	53.6	92.9	**100.0**
CL	2.302	2.180	**1.591**	5.631	**5.522**	7.664	3.966	**3.851**	4.628	46.8	46.9	**63.4**
Mean over cases	2.873	2.140	**1.787**	4.024	**3.297**	4.089	3.448	**2.719**	2.938	72.1	**78.4**	77.6

S4. Candidate-region feasibility

For each outer fold of the primary evaluation, the candidate regions of the held-out vertebrae were rebuilt from the training subjects exactly as in the method (Section S2), and each reference landmark of the held-out subject was assigned to the mesh vertex nearest to it. A landmark counts as inside if that vertex belongs to the region of its class. The GCZ selection step and the GCZ-RL projection step use the same bands but differ in their last fallback (Section S2); they produce identical regions for every class except ITL. By vertebral level, the percentage of reference landmarks inside their region was L1 87.5%, L2 96.9%, L3 95.1%, L4 89.5%, L5 76.2%.

Table S4: Candidate-region feasibility on the held-out vertebrae of the primary evaluation (14 vertebrae per class). Inside: reference landmarks whose nearest mesh vertex lies in the region of their class. Region size: median number of vertices (meshes have a median of 33,445 vertices). Fallback: cases in which fewer than 64 vertices satisfied all three bands. Side empty: ITL or CL cases in which the projection region had no vertex on one side of the mid-sagittal plane.

			Region size		Fallback		
Class	Ref.	Inside%	Selection	Projection	Selection	Projection	Side empty
ALL	368	96.7	6041	6041	0/14	0/14	0
PLL	239	83.7	2891	2891	0/14	0/14	0
LF	349	87.4	5254.5	5254.5	0/14	0/14	0
ISL	330	94.5	2764	2764	0/14	0/14	0
SSL	14	85.7	124.5	124.5	5/14	5/14	0
ITL	28	92.9	10990	34.5	14/14	14/14	2
CL	837	88.1	8467.5	8467.5	0/14	0/14	0
All	2165	90.0	–	–	19/98	19/98	2

The two side-empty cases are the two ITL vertebrae (S01 L5 and S02 L4) for which the corrected point keeps its GCZ position (Section 4.3).

S5. Implementation details and hyperparameters

Table S5 lists the settings of every component. All are fixed; none was tuned on held-out annotations except the per-class development configuration (Section 3.7), whose selection after inspection of held-out outputs is disclosed in the main text.

Table S5: Implementation settings.

Component	Setting
Candidate regions	$k_L = 2$; bands clipped to $[0, 1]$; fallback below 64 vertices (Section S2).
Atlas selection	ICP from every atlas vertebra, 1200 sampled atlas points, at most 40 iterations, tolerance 10^{-5}; residual = mean nearest-neighbor distance of 1200 sampled atlas points.
Full alignment	ICP, 5000 sampled points, at most 150 iterations, tolerance 10^{-6} (primary evaluation and development GCZ column) or 10^{-5} (development GCZ-RL input).

Component	Setting
GCZ selection	Snap radius 15 mm; left/right split for LF and ITL, four quadrants for CL; farthest-point sampling to ten landmarks. A fallback landmark template pooled from all training vertebrae is used only if the selected atlas lacks a class.
Residual targets	Hungarian pairing of GCZ and expert points; mirroring of LF, ISL, ITL, CL about the sagittal plane through the vertex centroid.
Correction bound	τ_L = mean + 2 SD of training residual magnitudes of class L (including mirrored samples).
Extra Trees ET-1	220 trees, maximum depth 18, minimum leaf size 1, 85% of features per split, no bootstrap, seed 8.
Extra Trees ET-2	520 trees, unlimited depth, minimum leaf size 1, all features, no bootstrap, seed 91.
Extra Trees ET-3	520 trees, maximum depth 13, minimum leaf size 2, 75% of features, no bootstrap, seed 92.
Extra Trees ET-4	520 trees, maximum depth 18, minimum leaf size 1, 85% of features, bootstrap, seed 93.
Other regressors (Table 3)	Ridge ($\lambda = 25$, standardized features); k-nearest neighbors ($k = \min(31, \max(5, \lfloor\sqrt{n}\rfloor))$, distance-weighted, standardized); random forest (420 trees, 85% of features, bootstrap, seed 94); gradient boosting (260 trees, learning rate 0.035, depth 3, subsample 0.85, seed 95) and histogram gradient boosting (260 iterations, learning rate 0.035, 31 leaves, L2 0.05, seed 96), each with one model per output coordinate.
Development configuration	ALL ET-2, $\alpha = 1.20$; PLL ET-2, 1.20; LF ET-1, 1.00; ISL ET-4, 1.20; SSL ET-1, 1.00; ITL ET-4, 1.00; CL ET-2, 1.10.
Primary selection grid	ET-1 to ET-4 $\times\ \alpha \in \{0, 0.25, 0.40, 0.55, 0.70, 0.85, 1.00, 1.10, 1.20\}$ per class and fold, inner leave-one-subject-out; ties broken by configuration name, then α (Table S9).
SLD	Released SpineLib code (commit 62f39cf); pyACVD remeshing (subdivision 3, 5000 clusters), 10 smoothing iterations, hole filling up to 1000, `Spine(max_angle=60)`, shape decomposition without lamina; landmarks snapped to the input mesh; LPS↔RAS conversion; centerline approximated by cross-section centroids.
Software	Python 3.11, NumPy 2.4, SciPy 1.17, pandas 3.0, scikit-learn 1.8, trimesh 4.12, VTK 9.6, PyVista 0.47, pyACVD 0.4.

Sensitivity of the atlas selection to the computing environment The reported results use the ICP alignments stored with the code release, from which every number is regenerated exactly. When all alignments of the primary evaluation were recomputed from scratch in a second computing environment (same code, same pinned library versions, and the same input files, verified by checksum), the selected atlas vertebra changed for 2 of the 14 held-out vertebrae (S01 L3 and S02 L3), and the changed training-side predictions altered the inner model selection of the S01 and S02 folds. The macro-averaged MAE became 3.208 mm for GCZ and 2.418 mm for GCZ-RL (Pass@5 83.1% and 89.8%); SLD, which does not use ICP, was unchanged at 2.224 mm. The S03 fold was unchanged. Most of the difference comes from S02 L3, for which the newly selected atlas had a lower quick-ICP residual (1.33 against 1.51 mm) but a GCZ error higher by 4.0 mm, that is, an alignment with a small surface distance but wrong correspondence. The improvement of GCZ-RL over GCZ (24.6%) is preserved, whereas the ordering of GCZ-RL and SLD by macro-averaged MAE is not. Atlas selection starts ICP from the scanner coordinates of each vertebra, and several candidate alignments converge to local minima with residuals of 4–5 mm; the selection is therefore sensitive to small numerical differences. A systematic analysis over ten ICP seeds is given in Section S13.

Correction of the calibration regressor comparison The leave-one-vertebra-out comparison of Table 3 evaluates, for each held-out calibration vertebra, the corrections predicted for its GCZ points. In the version reported in the first author's dissertation, the validation rows of the bilateral classes (LF, ISL, ITL, CL) were taken as the first n rows of that vertebra and class, where n is its number of GCZ points; because original and mirrored samples alternate, half of these rows were mirrored samples, and their predicted corrections were applied to the wrong GCZ points. In addition, zones, class prior, and correction bound were computed once from all 20 calibration vertebrae. Table S6 compares the published values with the corrected protocol used in Table 3, in which only original samples are validated, each with its own GCZ point, and zones, prior, and bound are recomputed without the held-out vertebra. The correction lowers the errors of LF, ISL, and CL for every family; for ITL the corrected errors are 0.16–0.44 mm higher, mainly because zones, class prior, and correction bound no longer include the held-out vertebra. Under the corrected protocol Extra Trees is the best family for every class;

the conclusion that Extra Trees is the most suitable residual regressor is unchanged. The development configuration had been fixed before the correction and is unchanged.

Table S6: Mean over classes of the leave-one-vertebra-out calibration MAE (mm) under the published and corrected protocols, and the MAE of the bilateral classes under the corrected protocol (published values in parentheses).

	Ridge	KNN	Random forest	Gradient boosting	Extra Trees
Mean, published protocol	3.103	3.017	2.754	2.777	2.575
Mean, validation rows corrected	2.986	2.655	2.241	2.274	1.984
Mean, rows and fold quantities corrected (Table 3)	3.055	2.715	2.307	2.337	2.037
LF	3.546 (3.760)	2.744 (3.653)	2.448 (3.329)	2.386 (3.543)	2.125 (3.354)
ISL	1.981 (2.013)	1.918 (1.945)	1.588 (1.831)	1.622 (1.919)	1.551 (1.805)
ITL	4.046 (3.816)	4.085 (3.833)	3.399 (3.097)	3.532 (3.090)	3.170 (3.014)
CL	4.817 (5.214)	3.700 (5.214)	2.934 (5.386)	2.963 (5.157)	2.578 (5.049)

S6. Diagnostic analysis of the correspondence rule

Pure Hungarian matching minimizes the total distance but can pair anatomically non-corresponding points, such as left–right swaps, superior–inferior inversions, and long jumps along curved attachment regions. To quantify this, an anatomy-constrained correspondence was evaluated for a geometry-only GCZ prediction on the nine held-out lumbar vertebrae of the development analysis with complete landmark sets (S02 L1–L4 and S03 L1–L5): ALL and PLL used side-aware matching with a superior–inferior band, LF and ITL strict left–right constraints, CL a four-quadrant strategy, and ISL and SSL local ordering and distance constraints; pairs that violated these rules were penalized or rejected.

Important difference from the main evaluation. In this diagnostic, which was run with the original analysis notebook, the number of GCZ landmarks per class was set to the number of reference points of the evaluated vertebra (at least four), so that nearly every reference point could be paired. This uses the cardinality, though not the positions, of the evaluated annotations. The analysis therefore characterizes the matching rules only. Its predictions are not those evaluated in the main text, which use a fixed number of at most ten landmarks per class, and its values are not comparable with the evaluation tables of the main text.

The nine vertebrae carry 1145 reference landmarks. Pure Hungarian matching paired 1126 of them (coverage 98.3%), whereas the constrained rules, which leave a landmark unpaired when no admissible candidate exists, paired 1039 (90.7%) and 1042 (91.0%). At this lower coverage, the final constrained rule had an MAE of 3.67 mm against 4.85 mm, an RMSE of 5.34 mm against 7.71 mm, a pass rate of 82.2% against 75.2%, and 58 instead of 119 errors above 10 mm (Table S7, Fig. S1). The error statistics of the three rows are computed over different sets of pairs and are not directly comparable, and part of the improvement may come from landmarks that the constrained rules leave unpaired. The analysis supports the narrower conclusion that anatomical constraints remove long-range mispairings. Table S8 reports the per-ligament MAE under the initial and final rules. The largest change was for ALL, from 5.07 to 4.17 mm, after side-aware and superior–inferior band constraints; PLL improved slightly. The other classes kept the initial rule, because further band, principal-axis, and ordering variants did not improve the balance between MAE, pass rate, and long errors.

Table S7: Pooled metrics of the diagnostic geometry-only GCZ prediction under three correspondence strategies (nine held-out lumbar vertebrae, 1145 reference landmarks). Coverage: paired reference landmarks as a percentage of all reference landmarks. Rows are computed over different sets of pairs. $N > 10$: pairs with an error above 10 mm.

Correspondence strategy	Pairs	Cov.%	MAE	SD	RMSE	Pass%	$N > 10$
Pure Euclidean Hungarian	1126	98.3	4.85	5.99	7.71	75.2	119
Initial anatomy-aware matching	1039	90.7	3.84	4.30	5.76	80.2	61
Final anatomy-constrained	1042	91.0	3.67	3.87	5.34	82.2	58

S7. Models selected in the primary evaluation

When the inner validation error was identical for several configurations, as for $\alpha = 0$, where the configuration does not matter, the first configuration in name order is listed.

Table S8: Per-ligament MAE (mm) of the diagnostic geometry-only GCZ prediction under the initial and final correspondence rules (nine held-out lumbar vertebrae).

Lig.	Initial MAE	Final MAE	$N > 10$
ALL	5.07	4.17	13
PLL	3.70	3.62	3
LF	3.21	3.21	14
ISL	4.43	4.43	15
SSL	1.86	1.86	0
ITL	2.68	2.68	0
CL	3.36	3.36	13

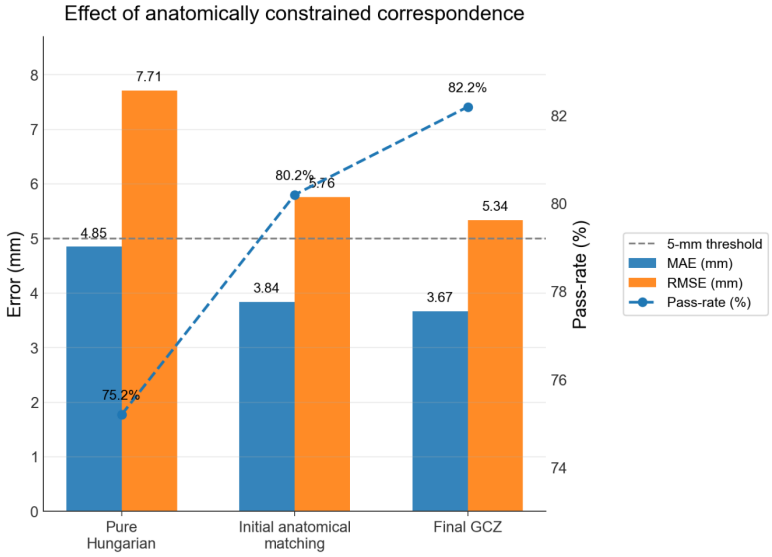


Figure S1: Effect of anatomically constrained correspondence on the diagnostic geometry-only GCZ prediction. Bars show pooled MAE and RMSE, and the dashed line shows the percentage of matched pairs within 5 mm (values of Table S7).

S8. Component ablation

The component analyses decompose the fixed pipeline of the primary evaluation; they were defined after the primary results were known and select nothing. They use the stored ICP geometry and the fold models retrained by `part2_patient_loso_14/run_part2.py` with their fixed seeds, and are computed by `paper/revision_analyses.py`. For each GCZ point, the residual is predicted by the Extra Trees configuration selected for its class in its fold (Table S9); where $\alpha = 0$ was selected, the first configuration in name order (ET-1) supplies the prediction for the intermediate configurations. The complete configuration reproduces the saved GCZ-RL predictions (maximum coordinate difference 6×10^{-14} mm, floating-point rounding). No predicted correction of a held-out point exceeded its bound τ_L, so the raw and bounded residuals coincide. Projecting the GCZ points without a residual changes only ITL, because the GCZ points of the other classes already lie in their projection regions.

The last row of Table S10 shows why the side constraint is part of the method. The complete nested procedure of the primary evaluation, including the inner selection of configuration and blend weight, was re-run with the constraint switched off. The inner selection was unchanged except for the ITL blend weight of the S02 fold (1.10 instead of 1.20), and every class except ITL kept its error. For ITL, both points of 2 of the 14 held-out vertebrae were then projected onto the same transverse process, an anatomically invalid configuration that the constraint excludes.

Against the strongest geometry-only configuration (GCZ with projection, 2.310 mm), the complete GCZ-RL reduced the macro-averaged MAE to 2.150 mm, and did so for each held-out subject (S01 2.563 to 2.315 mm, S02 2.042 to 1.986 mm, S03 2.273 to 2.117 mm; difference −0.160 mm, 95% interval −0.234 to −0.085 mm; Wilcoxon $p < 0.001$ over the 98 cases, 45 lower, 15 higher, 38 unchanged). The row with $\alpha_L = 1$ is a comparison made after the held-out evaluation; it was not used to choose the method, whose blend weights were selected by the nested inner validation. In the complete configuration, 740 of the 742 outputs are vertices of the projection region; the other two (ITL of S01 L5 and S02 L4) keep their GCZ position under the side constraint and are vertices of the GCZ selection region. Over the six classes other than ITL, where projection alone leaves the GCZ points unchanged, it reduced the macro-averaged MAE from 2.316 to 2.209 mm (difference −0.107 mm, 95% vertebra-level cluster-bootstrap interval −0.175 to −0.036 mm; Wilcoxon $p < 0.001$ over 84 vertebra–class cases, 33 lower, 13 higher, 38 unchanged), by 6.4%, 1.5%, and 4.6% for held-out S01,

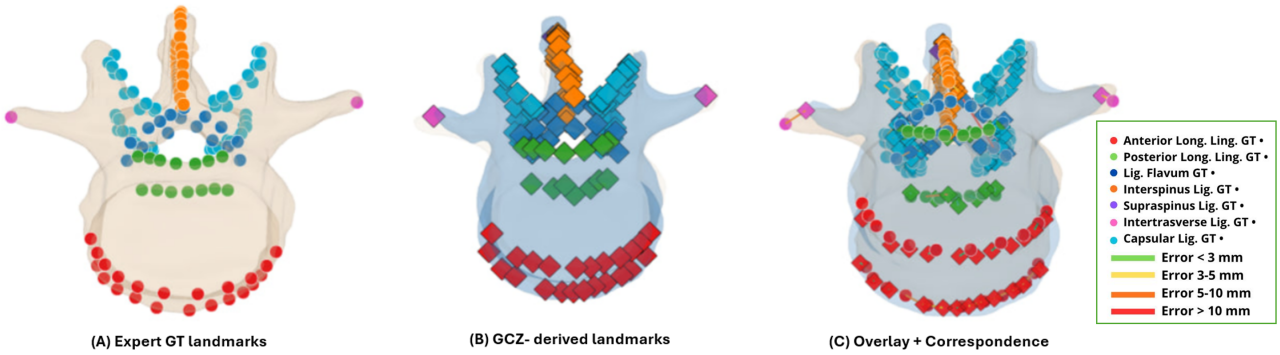


Figure S2: Representative qualitative landmark correspondence on a held-out vertebra of the diagnostic analysis. (A) Expert landmarks. (B) GCZ landmarks transferred from the atlas after alignment. (C) Overlay of expert and GCZ landmarks with correspondence lines; circles indicate expert landmarks and diamonds GCZ landmarks. The overlay exposes side swaps, superior–inferior inversions, and candidate-region failures.

Table S9: Extra Trees configuration and blend weight selected for each class by the inner leave-one-subject-out of each outer fold, with the inner validation MAE (mm). $\alpha = 0$ keeps the GCZ point.

	Held-out S01		Held-out S02		Held-out S03	
Class	Model, α	MAE	Model, α	MAE	Model, α	MAE
ALL	ET-4, 0.55	3.234	ET-3, 0.25	3.886	ET-1, 0	3.163
PLL	ET-1, 0	2.188	ET-1, 0	3.427	ET-1, 0	1.823
LF	ET-1, 0	4.482	ET-1, 0.40	1.873	ET-1, 0	2.195
ISL	ET-3, 0.70	4.603	ET-2, 0.70	2.325	ET-4, 0.85	2.106
SSL	ET-1, 0.25	2.745	ET-2, 0.40	3.309	ET-1, 0.25	3.516
ITL	ET-2, 1.20	2.218	ET-1, 1.20	1.694	ET-4, 1.20	1.130
CL	ET-2, 0.25	4.780	ET-1, 0	2.882	ET-1, 1.20	2.379

S02, and S03.

S9. Candidate-region coverage and compactness by held-out subject

For each held-out vertebra and class, the region used for projection was rebuilt from the training subjects of its fold. Its vertex fraction is the share of mesh vertices inside the region, and its area fraction is the share of the vertebral surface area that belongs to the region, where each vertex carries one third of the area of every incident triangle. Containment is computed as in Section S4. The GCZ selection step uses the same bands but a wider last fallback (Section S2); for ITL, which used this fallback in all 14 vertebrae, the selection region covers a median of 40.0% of the surface, against 0.1% for the projection region.

S10. Residual-learning interpretability

For every GCZ point of the held-out vertebrae, the true residual is the vector to its Hungarian-paired reference point (Eq. 7 of the main text), and the predicted residual is the output of the selected Extra Trees model before blending. Table S12 summarizes their agreement. Table S13 reports grouped permutation importance, the increase in the mean Euclidean error of the predicted residual on the held-out vertebrae when the features of one group are permuted jointly across points (30 repetitions per group and fold), and the impurity-based importance of the fitted models summed over the group. Importance describes what the fitted models use; it does not establish an anatomical or causal relation. Figure S3 shows the true and predicted residuals on one held-out vertebra.

By impurity, the ten most used single features were the position of the point in the calibrated y_n band (feature 9; 0.119), the superior–inferior and lateral components of the direction to the class prior (features 35 and 34; 0.080 and 0.057), the normalized lateral position (feature 1; 0.059), the signed lateral position (feature 32; 0.057), the distance to the region box boundary (feature 8; 0.051), the vertebral level (feature 20; 0.034), the distance from the mid-sagittal plane (feature 4; 0.034), and the normalized superior–inferior position and its repeat (features 3 and 6; 0.033 each). Grouped permutation importance ranked the groups differently in each fold (anatomical context first for held-out S01, normalized position for S02, local surface geometry for S03), so no single group dominates.

Table S10: Component ablation of the primary evaluation by ligament class: MAE (mm) over the 14 held-out vertebrae.

Configuration	ALL	PLL	LF	ISL	SSL	ITL	CL
GCZ (geometry only)	2.605	1.863	1.918	2.291	2.917	6.262	2.302
GCZ + projection, no residual	2.605	1.863	1.918	2.291	2.917	2.279	2.302
GCZ + raw residual	2.375	1.694	1.943	2.411	2.871	5.222	2.369
GCZ + bounded residual	2.375	1.694	1.943	2.411	2.871	5.222	2.369
GCZ + bounded residual + projection ($\alpha_L = 1$)	2.165	1.600	1.820	2.238	2.726	2.150	2.227
Complete GCZ-RL (nested α_L)	2.382	1.863	1.912	2.226	2.692	1.796	2.180
GCZ-RL without side constraint (nested re-run)	2.382	1.863	1.912	2.226	2.692	8.279	2.180

Table S11: Reference landmarks inside the projection region (%) and median area fraction of the region (%) by held-out subject.

	Held-out S01		Held-out S02		Held-out S03	
Class	Inside	Area	Inside	Area	Inside	Area
ALL	98.3	21.9	98.9	20.6	92.1	18.2
PLL	81.7	9.6	100.0	11.3	73.6	8.4
LF	84.7	19.0	93.8	17.8	86.7	16.4
ISL	93.9	8.2	98.1	7.6	91.4	8.4
SSL	80.0	0.3	100.0	0.4	80.0	0.5
ITL	90.0	0.1	87.5	0.1	100.0	0.1
CL	90.5	24.2	94.4	28.7	76.2	26.7

S11. Atlas selection and registration sensitivity

Table S14 lists, for each held-out vertebra of the primary evaluation, the atlas vertebra selected with the stored ICP alignments, whether it is of the same level, its quick-ICP residual, and the per-vertebra macro-averaged MAE of GCZ and GCZ-RL, together with the same quantities from the second computing environment (Section S5). The second environment is a recomputation on the same data, not an independent cohort. Same-level atlases were selected for 7 vertebrae and cross-level atlases for 7; mean per-vertebra GCZ errors were 3.073 and 2.686 mm, and mean GCZ-RL errors 2.203 and 2.098 mm. The quick-ICP residual was not associated with the per-vertebra error (Spearman $\rho = 0.24$, $p = 0.40$ for GCZ; $\rho = 0.25$, $p = 0.38$ for GCZ-RL). No new perturbation of the registration was performed for this analysis.

Table S12: Agreement of predicted and true residual vectors on the held-out vertebrae of the primary evaluation (all classes, whether or not the correction was applied). Clipped: after the correction bound. Cosine: cosine of the angle between predicted and true residual. Closer: share of points that the clipped correction alone moves closer to their paired reference point.

Class	Points	True (mm)	Clipped (mm)	Median cosine	Cosine > 0 (%)	Closer (%)
ALL	140	2.60	1.43	0.28	72.9	47.9
PLL	140	1.86	0.91	0.54	67.1	61.4
LF	140	1.92	0.91	0.33	66.4	53.6
ISL	140	2.29	1.11	0.26	63.6	45.7
SSL	14	2.92	2.59	0.75	85.7	42.9
ITL	28	6.26	3.78	0.47	82.1	53.6
CL	140	2.30	1.17	0.24	60.7	48.6

Table S13: Grouped feature importance of the residual regressors. Permutation: increase in held-out residual-prediction error, mean and range over the three folds. Impurity: share of the total impurity reduction, averaged over the fitted models of all folds. Groups (feature numbers of Table S1): normalized position 1–6 and 32; candidate-region relation 7–19; anatomical context 20–27; local surface geometry 28–31; offset from class prior 33–36.

Feature group	Permutation (mm)	Range over folds	Impurity share
Anatomical context (level, class)	0.200	0.038–0.378	0.077
Normalized position	0.189	0.110–0.332	0.255
Candidate-region relation	0.168	0.044–0.247	0.377
Local surface geometry	0.117	0.089–0.147	0.098
Offset from class prior	0.066	0.048–0.098	0.193

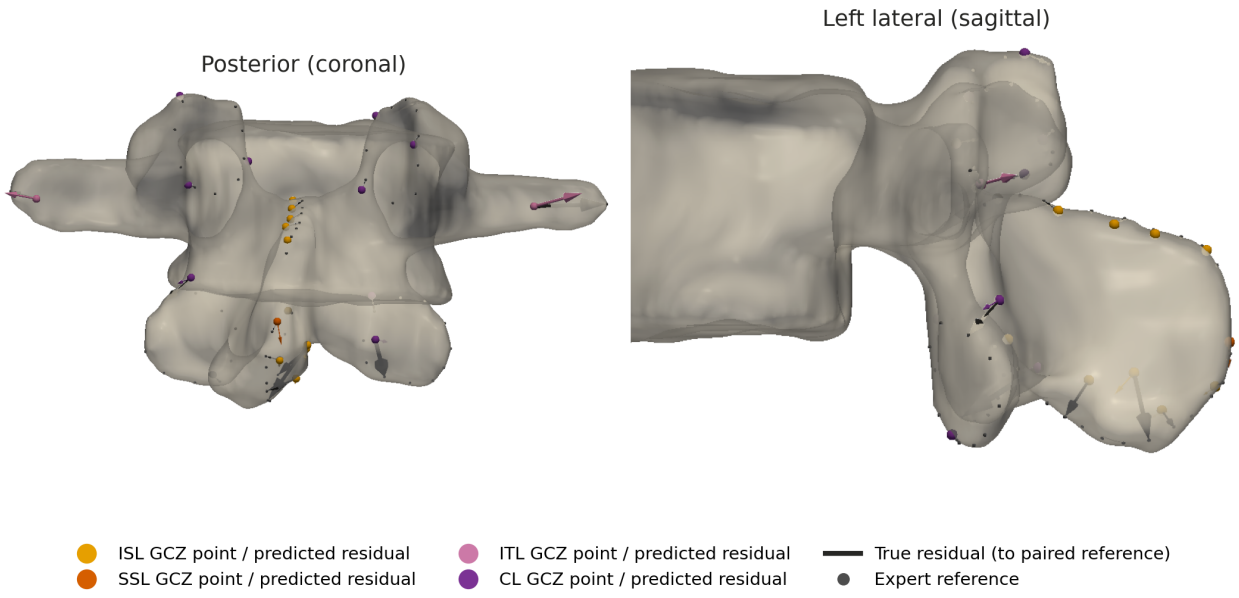


Figure S3: True (dark) and predicted (colored) residual vectors of the GCZ points on held-out vertebra S03 L4 (lumbar level L4; fold holding out S03), for the classes whose correction was applied in this fold (blend weight above zero): ISL, SSL, ITL, and CL. Small dark points are the expert reference landmarks of these classes. Vectors are drawn at true length, before the correction bound and projection.

Table S14: Atlas selection and per-vertebra error (mm) of the primary evaluation, with the stored ICP alignments and with the alignments recomputed in a second computing environment. *Atlas changed.

	Stored alignments (reported results)					Second environment			
Target	Atlas	Level	Residual	GCZ	GCZ-RL	Atlas	Residual	GCZ	GCZ-RL
S01 L1	S03 L1	same	1.38	4.360	2.466	S03 L1	1.38	4.360	2.536
S01 L2	S03 L2	same	1.56	3.470	2.089	S03 L2	1.56	3.470	1.971
S01 L3	S02 L2	cross	1.96	4.331	2.443	S03 L2*	1.74	4.938	2.459
S01 L4	S03 L4	same	1.73	3.287	1.834	S03 L4	1.73	3.287	1.714
S01 L5	S03 L5	same	1.75	3.475	2.743	S03 L5	1.75	3.475	2.774
S02 L1	S03 L2	cross	1.28	1.750	1.626	S03 L2	1.28	1.750	1.525
S02 L2	S03 L2	same	1.14	2.526	2.140	S03 L2	1.14	2.526	2.131
S02 L3	S01 L1	cross	1.51	2.286	2.234	S03 L3*	1.33	6.279	6.152
S02 L4	S03 L4	same	1.17	2.270	1.946	S03 L4	1.17	2.270	2.004
S03 L1	S01 L1	same	2.16	2.124	2.200	S01 L1	2.16	2.124	2.200
S03 L2	S01 L1	cross	1.61	2.251	2.002	S01 L1	1.61	2.251	2.002
S03 L3	S02 L1	cross	1.46	2.531	2.298	S02 L1	1.46	2.531	2.298
S03 L4	S01 L1	cross	1.64	3.263	1.972	S01 L1	1.64	3.263	1.972
S03 L5	S02 L4	cross	1.43	2.390	2.109	S02 L4	1.43	2.390	2.109

S12. Reproducibility verification

Table S15 compares the main reported values with the values regenerated from the public data by the code release (`paper/reproducibility_check.py`). Level A: metrics recomputed from the saved predictions and the annotations. Level B: end-to-end re-run with the stored ICP alignments and SLD output, with every Extra Trees model retrained with its fixed seed. Level C: ICP geometry recomputed from scratch; the leakage test was re-executed in this way, whereas the second-environment values of Section S5 are archived outputs that were not re-executed. The leakage test with recomputed ICP compared 412 points instead of 424, because the recomputed alignment selected a different atlas for S02 L3 (Section S11), and again found no difference. The S02 reduction of the submitted draft (10.1%) had been computed from rounded subject means; the unrounded value is 10.0%.

Table S15: Reproducibility verification of the main reported values.

Claim	Reported	Reproduced	$\lvert\Delta\rvert$	Level	Source file	Status
Primary cohort: vertebrae	14	14	0	B	`fold_details.json`	verified
Primary cohort: subjects	3	3	0	B	`fold_details.json`	verified
GCZ macro MAE (mm)	2.879	2.879	0	B	`table_P2_2_overall.csv`	verified
GCZ-RL macro MAE (mm)	2.150	2.150	0	B	`table_P2_2_overall.csv`	verified
SLD macro MAE (mm)	2.224	2.224	0	B	`table_P2_2_overall.csv`	verified
GCZ to GCZ-RL error reduction (%)	25.3	25.3	0	B	`table_P2_2_overall.csv`	verified
GCZ Pass@5 (%)	86.0	86.0	0	A	`paper_numbers.json`	verified
GCZ-RL Pass@5 (%)	92.7	92.7	0	A	`paper_numbers.json`	verified
SLD Pass@5 (%)	90.4	90.4	0	A	`paper_numbers.json`	verified
GCZ reference points within 5 mm (%)	72.1	72.1	0	A	`paper_numbers.json`	verified
GCZ-RL reference points within 5 mm (%)	78.4	78.4	0	A	`paper_numbers.json`	verified
SLD reference points within 5 mm (%)	77.6	77.6	0	A	`paper_numbers.json`	verified
GCZ-RL R-to-P closest point (mm)	3.297	3.297	0	A	`paper_numbers.json`	verified
SLD R-to-P closest point (mm)	4.089	4.089	0	A	`paper_numbers.json`	verified
GCZ R-to-P closest point (mm)	4.024	4.024	0	A	`paper_numbers.json`	verified
SLD P-to-R closest point (mm)	1.787	1.787	0	A	`paper_numbers.json`	verified
GCZ-RL P-to-R closest point (mm)	2.140	2.140	0	A	`paper_numbers.json`	verified
Reference landmarks inside candidate regions (%)	90.0	90.0	0	A	`paper_numbers.json`	verified
GCZ-RL minus GCZ (mm)	-0.729	-0.729	0	A	`paper_numbers.json`	verified
95% bootstrap CI, lower (mm)	-1.078	-1.078	0	A	`paper_numbers.json`	verified
95% bootstrap CI, upper (mm)	-0.381	-0.381	0	A	`paper_numbers.json`	verified
Wilcoxon p (reported as < 0.001)	<0.001	3.8e-06	0	A	`paper_numbers.json`	verified
GCZ-RL minus SLD (mm)	-0.074	-0.074	0	A	`paper_numbers.json`	verified
Wilcoxon p	0.35	0.35	0	A	`paper_numbers.json`	verified
Vertebrae improved by GCZ-RL over GCZ	13	13	0	A	`paper_numbers.json`	verified
S01: GCZ macro MAE (mm)	3.784	3.784	0	B	`table_P2_4_per_subject.csv`	verified
S01: GCZ-RL macro MAE (mm)	2.315	2.315	0	B	`table_P2_4_per_subject.csv`	verified
S01: reduction (%)	38.8	38.8	0	B	`per_vertebra_per_ligament_metrics.csv`	verified
S02: GCZ macro MAE (mm)	2.208	2.208	0	B	`table_P2_4_per_subject.csv`	verified
S02: GCZ-RL macro MAE (mm)	1.986	1.986	0	B	`table_P2_4_per_subject.csv`	verified
S02: reduction (%)	10.1	10.0	0.1	B	`per_vertebra_per_ligament_metrics.csv`	corrected to 10.0 (rounding)
S03: GCZ macro MAE (mm)	2.512	2.512	0	B	`table_P2_4_per_subject.csv`	verified
S03: GCZ-RL macro MAE (mm)	2.117	2.117	0	B	`table_P2_4_per_subject.csv`	verified
S03: reduction (%)	15.7	15.7	0	B	`per_vertebra_per_ligament_metrics.csv`	verified
ALL: GCZ MAE (mm)	2.605	2.605	0	B	`table_P2_3_per_ligament.csv`	verified
ALL: SLD MAE (mm)	1.658	1.658	0	B	`table_P2_3_per_ligament.csv`	verified
ALL: GCZ-RL MAE (mm)	2.382	2.382	0	B	`table_P2_3_per_ligament.csv`	verified
PLL: GCZ MAE (mm)	1.863	1.863	0	B	`table_P2_3_per_ligament.csv`	verified
PLL: SLD MAE (mm)	2.445	2.445	0	B	`table_P2_3_per_ligament.csv`	verified
PLL: GCZ-RL MAE (mm)	1.863	1.863	0	B	`table_P2_3_per_ligament.csv`	verified

Claim	Reported	Reproduced	$\lvert\Delta\rvert$	Level	Source file	Status
LF: GCZ MAE (mm)	1.918	1.918	0	B	table_P2_3_per_ligament.csv	verified
LF: SLD MAE (mm)	4.997	4.997	0	B	table_P2_3_per_ligament.csv	verified
LF: GCZ-RL MAE (mm)	1.912	1.912	0	B	table_P2_3_per_ligament.csv	verified
ISL: GCZ MAE (mm)	2.291	2.291	0	B	table_P2_3_per_ligament.csv	verified
ISL: SLD MAE (mm)	2.039	2.039	0	B	table_P2_3_per_ligament.csv	verified
ISL: GCZ-RL MAE (mm)	2.226	2.226	0	B	table_P2_3_per_ligament.csv	verified
SSL: GCZ MAE (mm)	2.917	2.917	0	B	table_P2_3_per_ligament.csv	verified
SSL: SLD MAE (mm)	1.968	1.968	0	B	table_P2_3_per_ligament.csv	verified
SSL: GCZ-RL MAE (mm)	2.692	2.692	0	B	table_P2_3_per_ligament.csv	verified
ITL: GCZ MAE (mm)	6.262	6.262	0	B	table_P2_3_per_ligament.csv	verified
ITL: SLD MAE (mm)	0.873	0.873	0	B	table_P2_3_per_ligament.csv	verified
ITL: GCZ-RL MAE (mm)	1.796	1.796	0	B	table_P2_3_per_ligament.csv	verified
CL: GCZ MAE (mm)	2.302	2.302	0	B	table_P2_3_per_ligament.csv	verified
CL: SLD MAE (mm)	1.591	1.591	0	B	table_P2_3_per_ligament.csv	verified
CL: GCZ-RL MAE (mm)	2.180	2.180	0	B	table_P2_3_per_ligament.csv	verified
Leakage test: max coordinate difference (mm)	0.000	0.000	0	C	table_P2_5_leakage_test.json	verified
Leakage test: points compared	424	424	0	C	table_P2_5_leakage_test.json	verified
Development: GCZ macro MAE (mm)	3.708	3.708	0	B	table_4_4_heldout_results.csv	verified
Development: GCZ Pass@5 (%)	79.8	79.8	0	B	table_4_4_heldout_results.csv	verified
Development: GCZ-RL macro MAE (mm)	2.946	2.946	0	B	table_4_4_heldout_results.csv	verified
Development: GCZ-RL Pass@5 (%)	88.1	88.1	0	B	table_4_4_heldout_results.csv	verified
Calibration comparison: Extra trees mean MAE (mm)	2.037	2.037	0	B	table3_rows_fixed_ctx.csv	verified
Calibration comparison: Random forest mean MAE (mm)	2.307	2.307	0	B	table3_rows_fixed_ctx.csv	verified
Calibration comparison: Gradient boosting mean MAE (mm)	2.337	2.337	0	B	table3_rows_fixed_ctx.csv	verified
Second environment: GCZ macro MAE (mm)	3.208	3.208	0	C (archived)	sensitivity_fresh_icp/table_P2_2_overall.csv	verified
Second environment: GCZ-RL macro MAE (mm)	2.418	2.418	0	C (archived)	sensitivity_fresh_icp/table_P2_2_overall.csv	verified
Recomputed ICP, seed offset 0: GCZ macro MAE (mm)	3.208	3.208	0	C	icp_seed_sensitivity/offset_0.json	verified
Recomputed ICP, seed offset 0: GCZ-RL macro MAE (mm)	2.418	2.418	0.0002	C	icp_seed_sensitivity/offset_0.json	verified

S13. Sensitivity to the ICP seeds

The atlas selection and alignment of GCZ use random point samples in ICP (Section S5). To quantify how much the results depend on this step, the complete nested procedure of the primary evaluation was re-run ten times with the ICP geometry recomputed from scratch, with every random seed of the ICP point sampling (quick ICP of each candidate atlas, residual sample, full alignment) shifted by a fixed offset per run (`paper/icp_seed_sensitivity.py`). Everything else was unchanged: the three LOSO folds and their subjects, the calibrated regions (computed from annotations only), the fallback template and farthest-point sampling seeds, the Extra Trees configurations and seeds, the model and blend-weight grids, the nested inner selection, and the evaluation. The residual training data change only as a consequence of ICP, because the training-side GCZ predictions are also recomputed. The stored checkpoints were neither read nor written. Offset 0 uses the seeds of the method and reproduces the second-environment run of Section S5 exactly (GCZ 3.208, GCZ-RL 2.418 mm). The ten runs are a computational sensitivity analysis on the same three subjects, not additional subjects or an

external validation.

Across the ten runs, 2–4 of the 14 held-out vertebrae changed atlas relative to the stored alignments, most often at L3 (S02 L3 in 8 runs). GCZ-RL improved on GCZ in every run, by 22.8–33.4% (mean 26.9%), and for every held-out subject in every run (30 of 30 comparisons). Its macro-averaged MAE ranged from 2.103 to 2.918 mm (mean 2.357 ± 0.239 mm), against 2.844 to 3.888 mm for GCZ (mean 3.226 ± 0.309 mm). SLD, which does not use ICP, is 2.224 mm in every run; GCZ-RL was below it in 3 of the 10 runs. The reported results (stored alignments: 2.150 and 2.879 mm) lie in the lower part of these ranges.

Table S16: Primary evaluation with the ICP alignments recomputed for ten seed offsets. MAE in mm (macro average over classes); reduction of GCZ-RL relative to GCZ; Pass@5 of GCZ-RL (%, averaged over vertebra–class cases); GCZ-RL MAE by held-out subject; held-out subjects improved by GCZ-RL over GCZ; held-out vertebrae whose atlas differs from the stored alignments.

Offset	GCZ	GCZ-RL	Reduction (%)	Pass@5	S01	S02	S03	Subjects improved	Atlas changed
0	3.208	2.418	24.6	89.4	2.291	2.953	2.117	3/3	2/14
1	3.425	2.459	28.2	89.4	2.408	3.005	2.073	3/3	4/14
2	2.966	2.103	29.1	94.9	2.136	2.056	2.108	3/3	2/14
3	3.467	2.310	33.4	92.3	2.784	2.027	2.060	3/3	3/14
4	3.260	2.379	27.0	89.6	2.331	2.857	2.044	3/3	3/14
5	2.844	2.134	25.0	93.0	2.327	1.988	2.057	3/3	2/14
6	3.223	2.433	24.5	89.4	2.302	3.043	2.075	3/3	2/14
7	2.986	2.306	22.8	90.9	2.280	2.631	2.073	3/3	3/14
8	3.888	2.918	25.0	84.7	2.481	2.979	3.306	3/3	3/14
9	2.995	2.114	29.4	94.2	2.154	2.114	2.074	3/3	2/14
Mean ± SD	3.226 ± 0.309	2.357 ± 0.239	26.9	90.8	2.349	2.565	2.199	10/10 runs	2.6

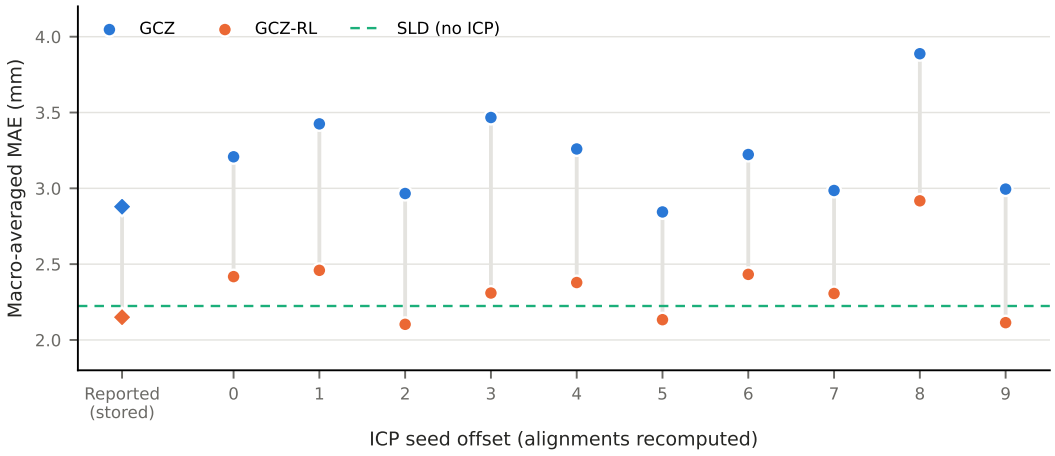


Figure S4: Macro-averaged MAE of GCZ (blue) and GCZ-RL (orange) for the reported run with the stored alignments (diamonds) and for ten runs with the ICP alignments recomputed with shifted seeds (circles); each gray segment connects the two methods of one run. Dashed line: SLD, which does not use ICP.

References

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